Phase2/Week4/Day1 – Retrieval Pipeline

Class Assignment:

Steps:

1. Unzip the project folder.

2. Open the project in Visual Studio Code.

3. Open the terminal from the project root folder and run:

npm install

Create the collection and vector index for userstory
Index definition file is available in
src/config/user-stories-vector-index.json

Update the .env file with mongodb credentials and mistral api key

4. After all packages are installed, start the application using:

npm run dev

5. The UI will open in the browser at:

http://localhost:3000

6. In the application, select:

Convert to JSON

7. Upload the test case //userstories Excel file available in:

src\data

8. After the JSON conversion is completed, select:

Embeddings & Store

9. Upload the converted JSON file and click:

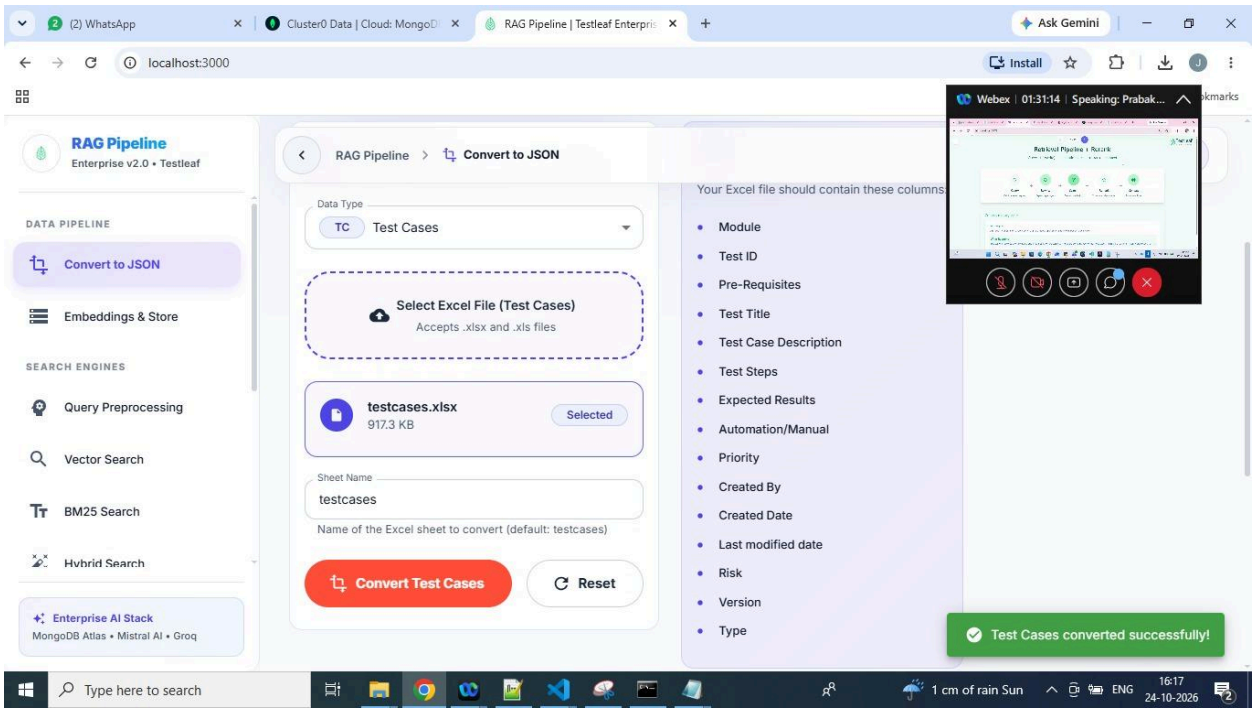
Create Embeddings

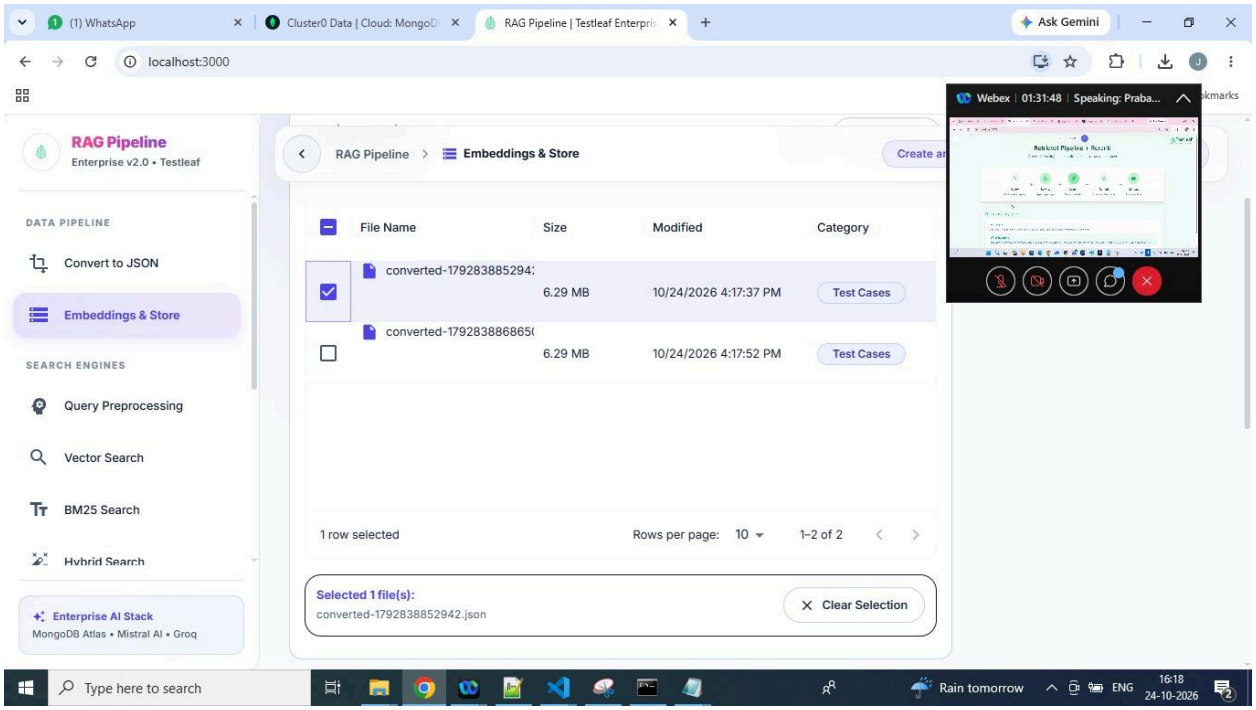
10. Monitor the ingestion process in:

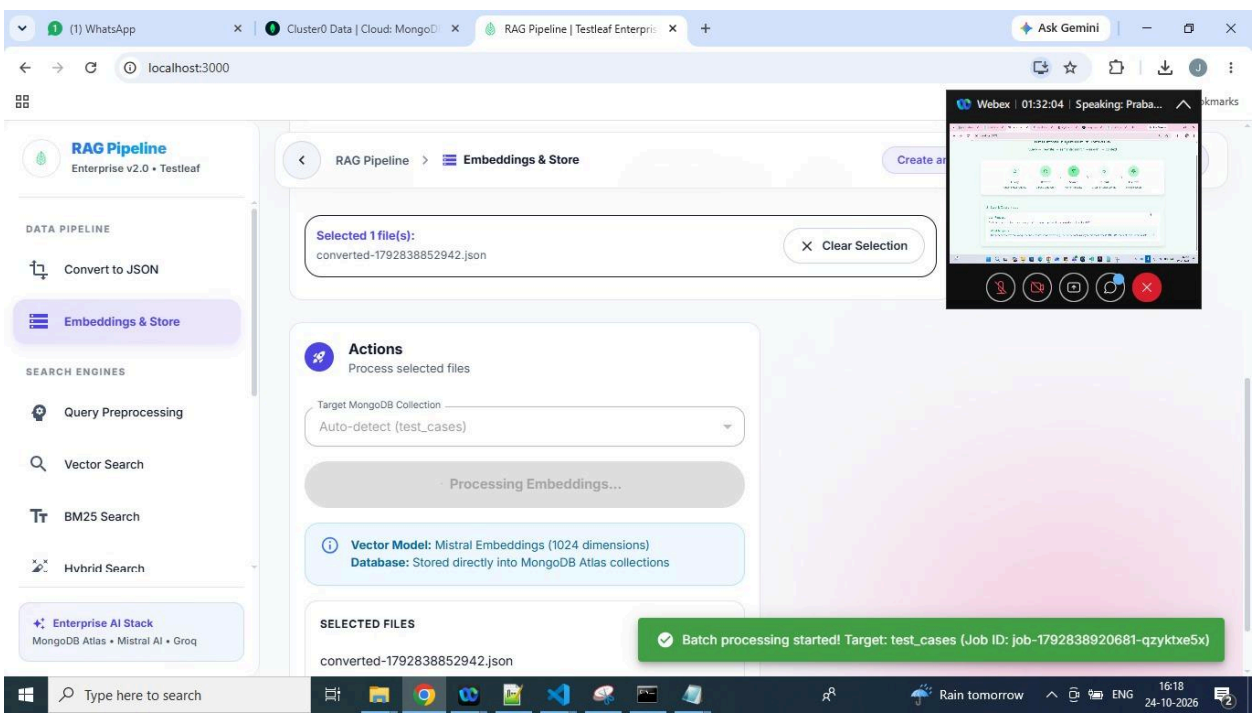
*** VS Code terminal logs**

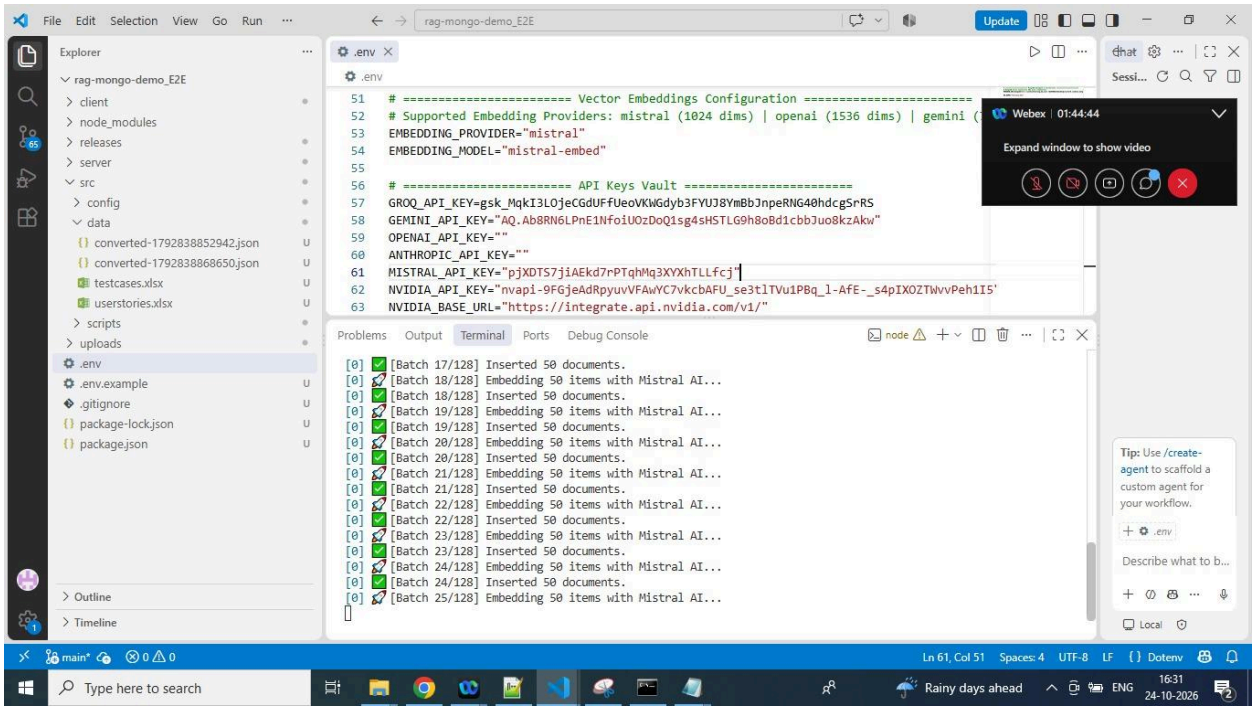
*** MongoDB database collections**

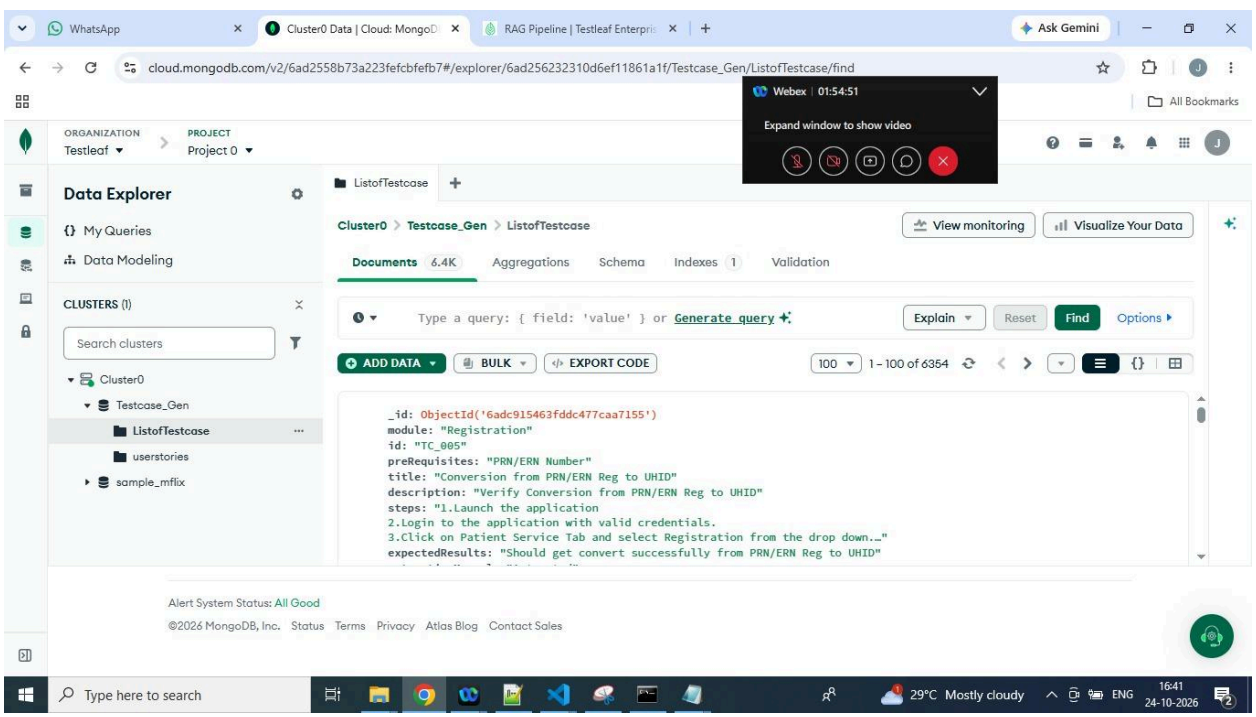
11. Wait until all 6354 test cases //190 user stories are successfully ingested.



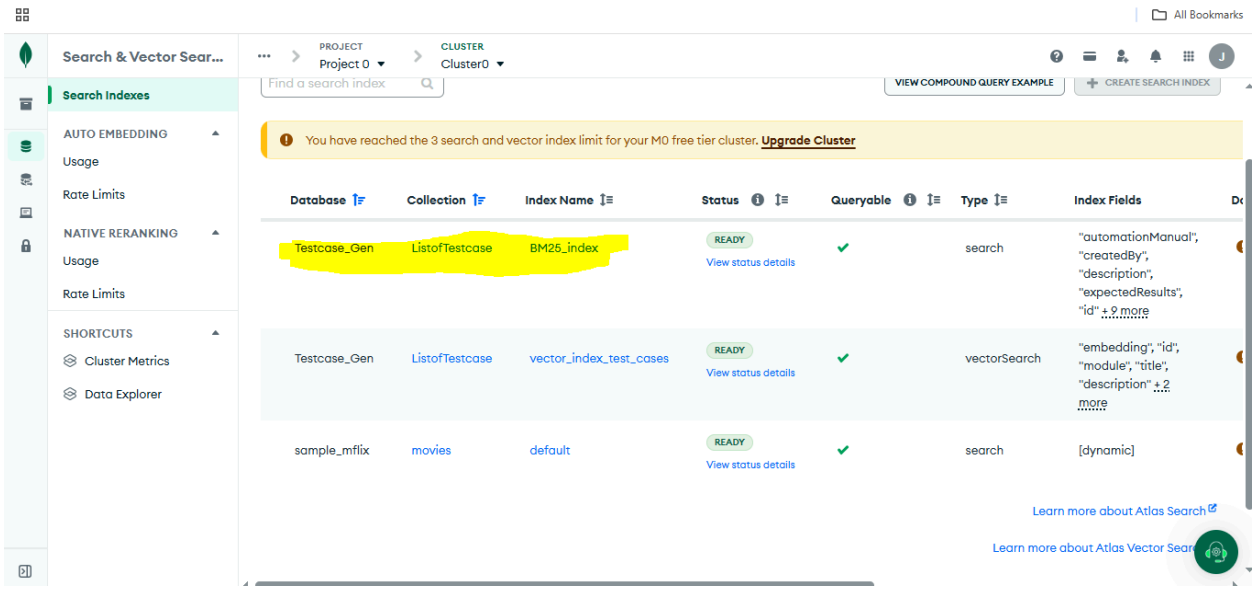

(1) WhatsApp
Cluster0 Data | Cloud: MongoD
RAG Pipeline | Testleaf Enterpris
Ask Gemini
localhost:3000
Webex | 01:31:48 | Speaking: Praba...
RAG Pipeline
Enterprise v2.0 • Testleaf
DATA PIPELINE
Convert to JSON
Embeddings & Store
SEARCH ENGINES
Query Preprocessing
Vector Search
BM25 Search
Hybrid Search
Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq
RAG Pipeline
Embeddings & Store
File Name
Size
Modified
Category
converted-179283885294
6.29 MB
10/24/2026 4:17:37 PM
Test Cases
converted-1792838868650
6.29 MB
10/24/2026 4:17:52 PM
Test Cases
1 row selected
Rows per page: 10
1–2 of 2
Selected 1 file(s):
converted-1792838852942.json
Clear Selection
Type here to search
Rain tomorrow
ENG
16:18
24-10-2026


(1) WhatsApp
Cluster0 Data | Cloud: MongoD
RAG Pipeline | Testleaf Enterpris
Ask Gemini
localhost:3000
Webex | 01:32:04 | Speaking: Praba...
RAG Pipeline
Enterprise v2.0 • Testleaf
DATA PIPELINE
Convert to JSON
Embeddings & Store
SEARCH ENGINES
Query Preprocessing
Vector Search
BM25 Search
Hybrid Search
Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq
RAG Pipeline
Embeddings & Store
Selected 1 file(s):
converted-1792838852942.json
Clear Selection
Actions
Process selected files
Target MongoDB Collection
Auto-detect (test_cases)
Processing Embeddings...
Vector Model: Mistral Embeddings (1024 dimensions)
Database: Stored directly into MongoDB Atlas collections
SELECTED FILES
converted-1792838852942.json
Batch processing started! Target: test_cases (Job ID: job-1792838920681-qzyktxe5x)
Type here to search
Rain tomorrow
ENG
16:18
24-10-2026


File Edit Selection View Go Run
rag-mongo-demo_E2E
Update
Explorer
rag-mongo-demo_E2E
client
node_modules
releases
server
src
config
data
converted-1792838852942.json
converted-1792838868650.json
testcases.xlsx
userstories.xlsx
scripts
uploads
.env
.env.example
.gitignore
package-lock.json
package.json
Outline
Timeline
.env
.env
51 # ========== Vector Embeddings Configuration ==========
52 # Supported Embedding Providers: mistral (1024 dims) | openai (1536 dims) | gemini (
53 EMBEDDING_PROVIDER="mistral"
54 EMBEDDING_MODEL="mistral-embed"
55
56 # ========== API Keys Vault ==========
57 GROQ_API_KEY=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
58 GEMINI_API_KEY="AQ.Ab8RN6LPnE1NfoiUOzDoQ1sg4sHSTLG9h8oBd1cbbJuo8kzAkw"
59 OPENAI_API_KEY=""
60 ANTHROPIC_API_KEY=""
61 MISTRAL_API_KEY="pjXDTS7jiAEkd7rPTqhMq3XYXhTLLfcj"
62 NVIDIA_API_KEY="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"
63 NVIDIA_BASE_URL="https://integrate.api.nvidia.com/v1/"
Problems Output Terminal Ports Debug Console
node
[0] [Batch 17/128] Inserted 50 documents.
[0] [Batch 18/128] Embedding 50 items with Mistral AI...
[0] [Batch 18/128] Inserted 50 documents.
[0] [Batch 19/128] Embedding 50 items with Mistral AI...
[0] [Batch 19/128] Inserted 50 documents.
[0] [Batch 20/128] Embedding 50 items with Mistral AI...
[0] [Batch 20/128] Inserted 50 documents.
[0] [Batch 21/128] Embedding 50 items with Mistral AI...
[0] [Batch 21/128] Inserted 50 documents.
[0] [Batch 22/128] Embedding 50 items with Mistral AI...
[0] [Batch 22/128] Inserted 50 documents.
[0] [Batch 23/128] Embedding 50 items with Mistral AI...
[0] [Batch 23/128] Inserted 50 documents.
[0] [Batch 24/128] Embedding 50 items with Mistral AI...
[0] [Batch 24/128] Inserted 50 documents.
[0] [Batch 25/128] Embedding 50 items with Mistral AI...
Webex | 01:44:44
Expand window to show video
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Describe what to b...
Local
main*
Ln 61, Col 51 Spaces: 4 UTF-8 LF Dotenv
Type here to search
Rainy days ahead
ENG
16:31
24-10-2026


WhatsApp
Cluster0 Data | Cloud: MongoD
RAG Pipeline | Testleaf Enterpris
Ask Gemini
cloud.mongodb.com/v2/6ad2558b73a223fefcbfefb7#/explorer/6ad256232310d6ef11861a1f/Testcase_Gen/ListofTestcase/find
Webex | 01:54:51
Expand window to show video
All Bookmarks
ORGANIZATION
Testleaf
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
Testcase_Gen
ListofTestcase
userstories
sample_mflix
ListofTestcase
Cluster0 > Testcase_Gen > ListofTestcase
View monitoring
Visualize Your Data
Documents 6.4K
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
100
1 – 100 of 6354
_id: ObjectId('6adc915463fddc477caa7155')
module: "Registration"
id: "TC_005"
preRequisites: "PRN/ERN Number"
title: "Conversion from PRN/ERN Reg to UHID"
description: "Verify Conversion from PRN/ERN Reg to UHID"
steps: "1.Launch the application
2.Login to the application with valid credentials.
3.Click on Patient Service Tab and select Registration from the drop down..."
expectedResults: "Should get convert successfully from PRN/ERN Reg to UHID"
Alert System Status: All Good
©2026 MongoDB, Inc. Status Terms Privacy Atlas Blog Contact Sales
Type here to search
29°C Mostly cloudy
ENG
16:41
24-10-2026

Home Assignment : Create a BM25 Index



Create a BM25 Index

Go to search & vector search

Click Create search index

Select atlas search

select the respective database and collection (testcases)

Click Json editor

paste the index definition below

```
{
  "mappings": {
    "dynamic": false,
    "fields": {
      "id": {
```

```
  "type": "string",
  "analyzer": "lucene.keyword"
},
"module": {
  "type": "string",
  "analyzer": "lucene.standard"
},
"title": {
  "type": "string",
  "analyzer": "lucene.standard"
},
"description": {
  "type": "string",
  "analyzer": "lucene.standard"
},
"steps": {
  "type": "string",
  "analyzer": "lucene.standard"
},
"expectedResults": {
  "type": "string",
  "analyzer": "lucene.standard"
},
```

```
"preRequisites": {
  "type": "string",
  "analyzer": "lucene.standard"
},
"automationManual": {
  "type": "string",
  "analyzer": "lucene.keyword"
},
"priority": {
  "type": "string",
  "analyzer": "lucene.keyword"
},
"risk": {
  "type": "string",
  "analyzer": "lucene.keyword"
},
"type": {
  "type": "string",
  "analyzer": "lucene.keyword"
},
"linkedStories": {
  "type": "string",
  "analyzer": "lucene.keyword"
```

```
      },
      "createdBy": {
        "type": "string",
        "analyzer": "lucene.keyword"
      },
      "version": {
        "type": "string",
        "analyzer": "lucene.keyword"
      }
    }
  }
}
```

click next and create search index

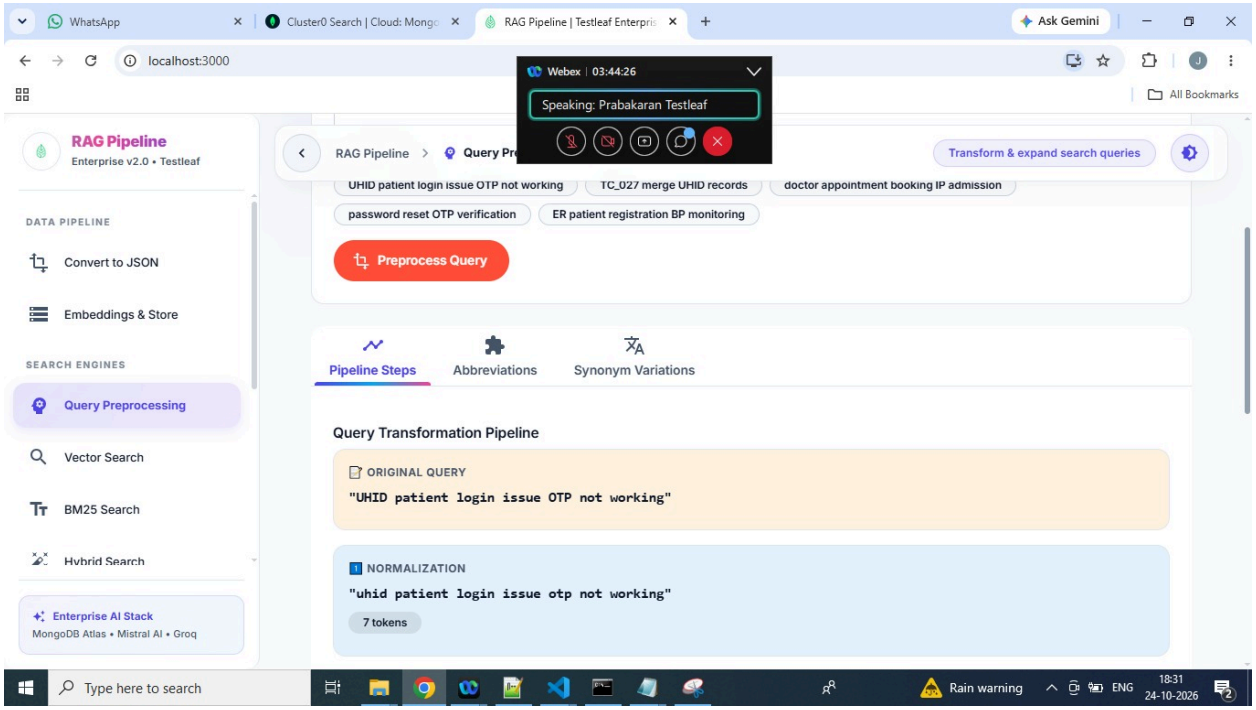
Update .env file

npm install

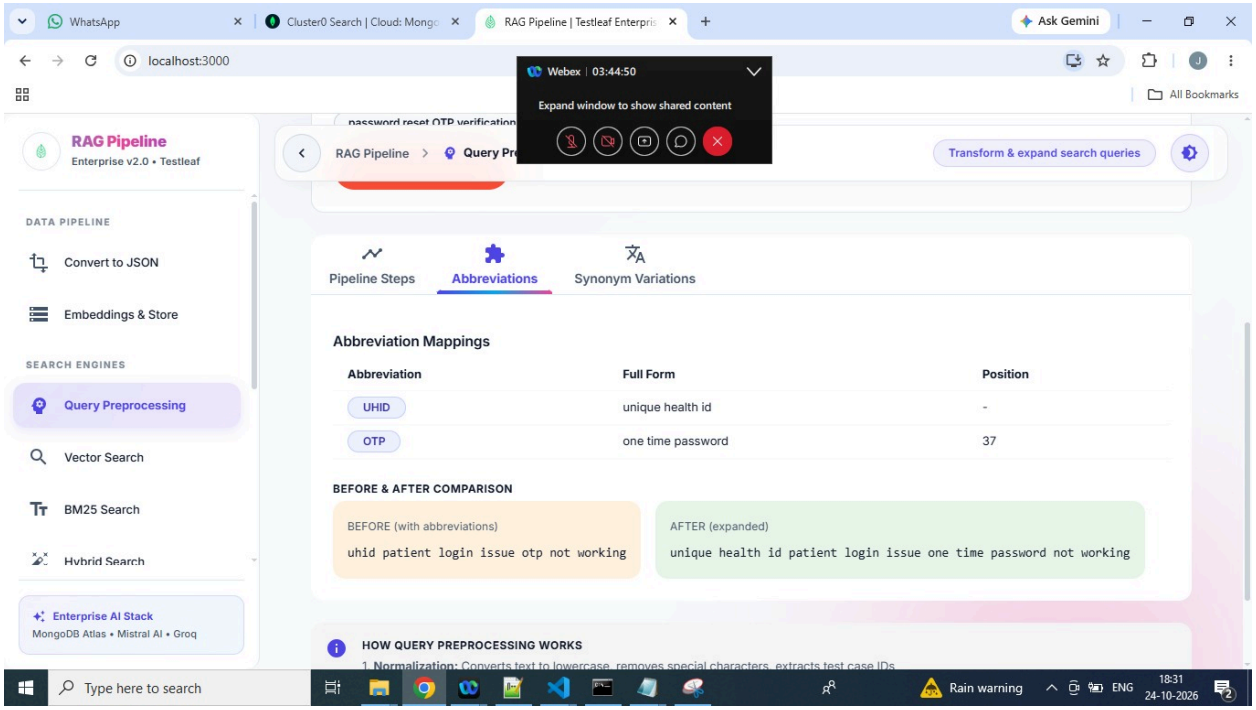
npm run dev

explore till hybrid search

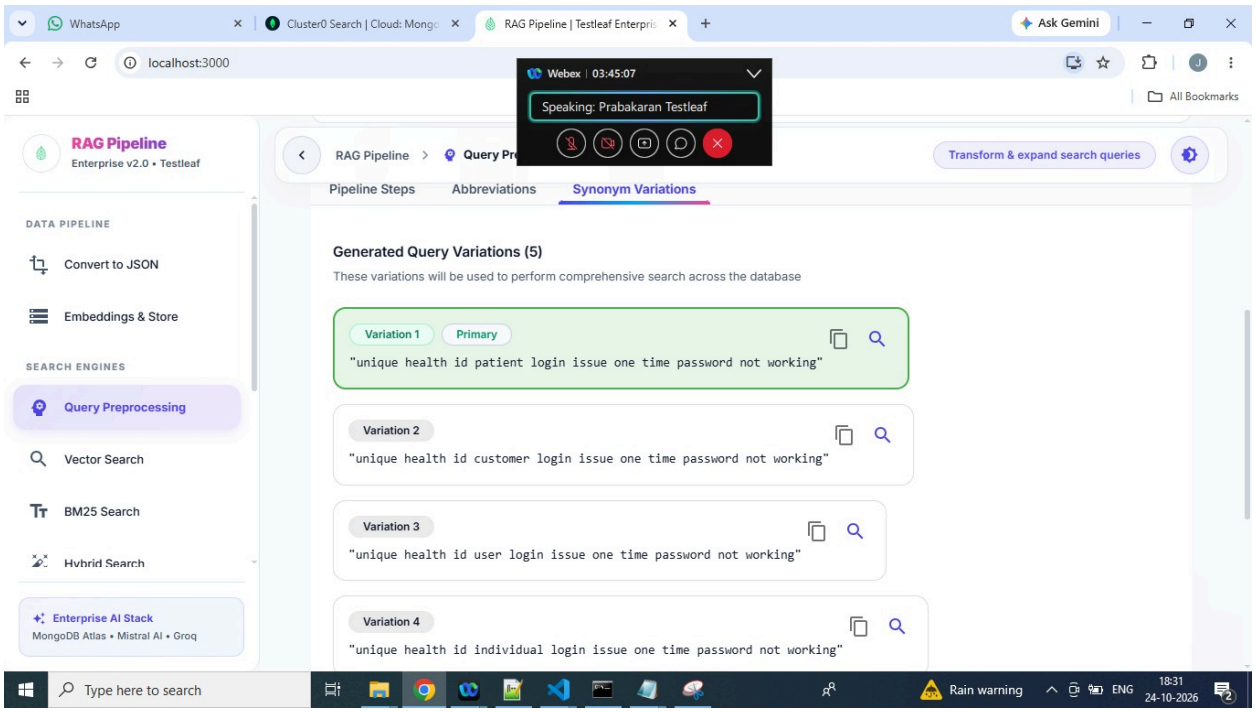
Query Processing:
Pipeline Steps:



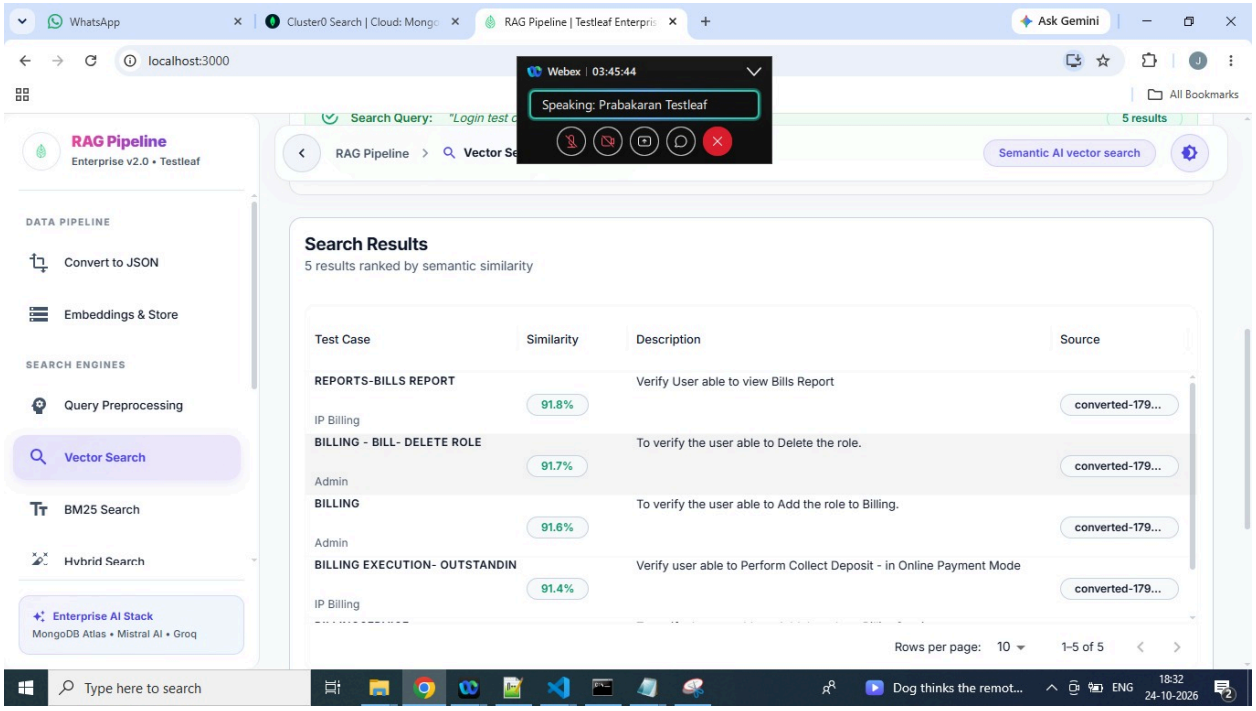
Abbreviations:



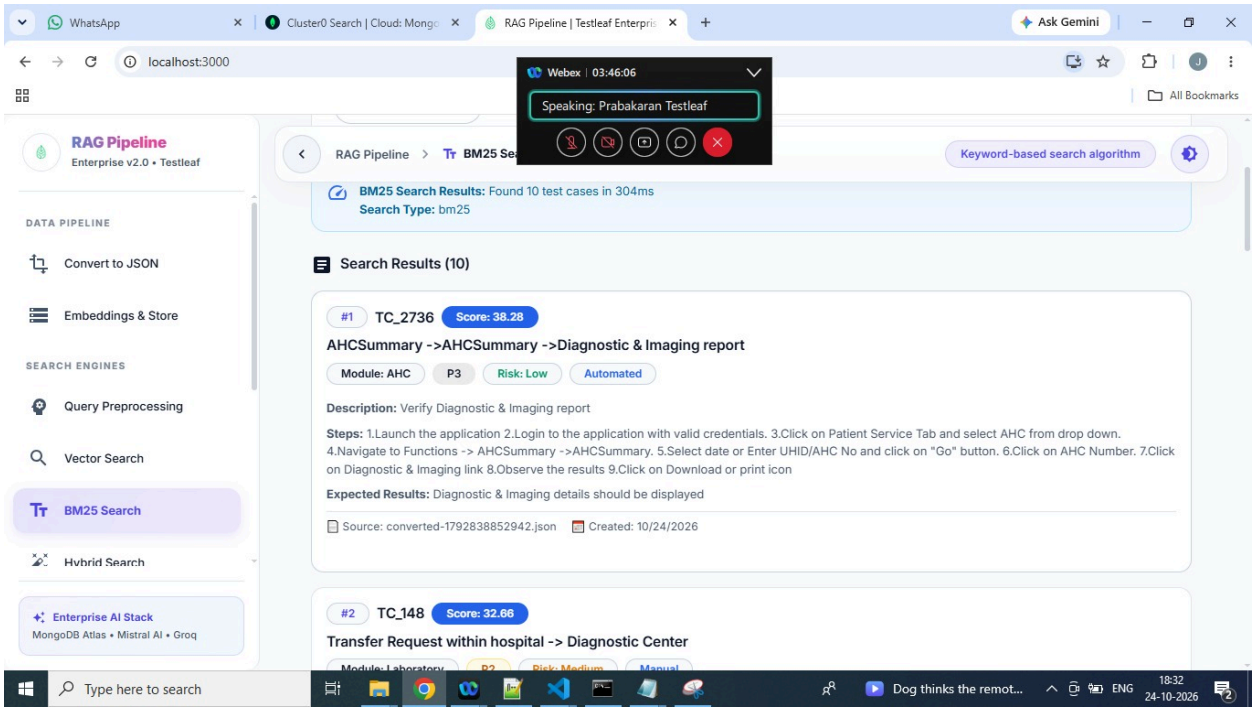
Synonym Variations:



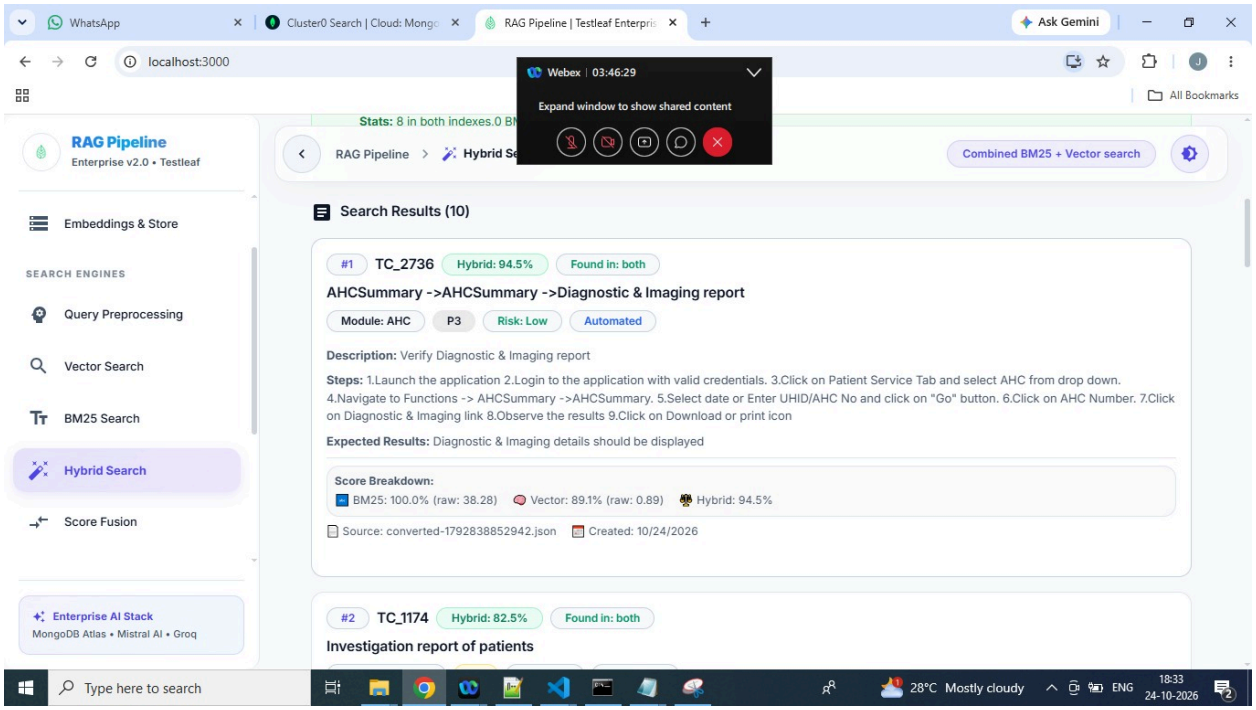
Vector Search :



BM25 Search:



Hybrid Search:


WhatsApp
Cluster0 Search | Cloud: Mongo
RAG Pipeline | Testleaf Enterpris
Ask Gemini
localhost:3000
All Bookmarks
Webex | 03:46:29
Expand window to show shared content
RAG Pipeline
Enterprise v2.0 • Testleaf
Embeddings & Store
SEARCH ENGINES
Query Preprocessing
Vector Search
BM25 Search
Hybrid Search
Score Fusion
Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq
RAG Pipeline
Combined BM25 + Vector search
Search Results (10)
#1
TC_2736
Hybrid: 94.5%
Found in: both
AHCSummary ->AHCSummary ->Diagnostic & Imaging report
Module: AHC
P3
Risk: Low
Automated
Description: Verify Diagnostic & Imaging report
Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select AHC from drop down. 4.Navigate to Functions -> AHCSummary ->AHCSummary. 5.Select date or Enter UHID/AHC No and click on "Go" button. 6.Click on AHC Number. 7.Click on Diagnostic & Imaging link 8.Observe the results 9.Click on Download or print icon
Expected Results: Diagnostic & Imaging details should be displayed
Score Breakdown:
BM25: 100.0% (raw: 38.28)
Vector: 89.1% (raw: 0.89)
Hybrid: 94.5%
Source: converted-1792838852942.json
Created: 10/24/2026
#2
TC_1174
Hybrid: 82.5%
Found in: both
Investigation report of patients
Type here to search
28°C Mostly cloudy
ENG
18:33
24-10-2026